CR23532 – CYBER AND DIGITAL FORENSICS

CASE STUDY ASSIGNMENTS – II

(Assignments 2(1) to 2(5))

A MINI PROJECT REPORT

submitted in partial fulfilment of the course requirements

Submitted by

DANIEL JOSHUA C S

Register Number: 241901020

B.E. COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)

Thandalam, Chennai – 602 105

OCTOBER 2026

TABLE OF CONTENTS

No.	Title	Page
–	Summary of Unique Scenarios Selected	3
1	Assignment 2(1) – Unauthorised Access to the CSE Examination Server	4
2	Assignment 2(2) – Linking “GradeSync.exe” to a User on a Shared Placement-Cell PC	11
3	Assignment 2(3) – Imaging and Safekeeping of a Research-Lab Laptop and USB Drive	17
4	Assignment 2(4) – Recovering Deleted Purchase-Office Files from Volume Shadow Copies	23
5	Assignment 2(5) – Multi-Source Investigation of a Compromised LMS Quiz Account	28
–	Conclusion and References	34

List of Figures

Fig.	Description
1.1	Forensic investigation flowchart – complaint to presentation (Assignment 2(1))
2.1	ILookIX analysis workflow (Assignment 2(2))
3.1	Seizure and acquisition flowchart using IXImager (Assignment 2(3))
4.1	Volume Shadow Copy recovery workflow (Assignment 2(4))
5.1	Multi-source evidence map (Assignment 2(5))

Summary of Unique Scenarios Selected

The assignment brief requires every student to frame a unique scenario for each unit. The following scenarios were created by the author for this submission. All persons, accounts, serial numbers, IP addresses and timestamps are fictitious. All times are recorded in **Indian Standard Time (IST, UTC+05:30)** unless stated otherwise.

Assign.	Unique Scenario Title	Case Reference	Core Concepts Applied
2(1)	"Operation Midnight Paper" – Unauthorised night-time copying of semester question papers from the CSE Department examination server EXAMSRV-CSE02	REC-CDF-2026-0315	Investigation phases, evidence classification, preservation, timeline, reporting, legal & ethical issues
2(2)	"The GradeSync Case" – Attributing a suspicious executable on the shared Training & Placement Cell desktop PC-TPO-04 to one of four user accounts	REC-CDF-2026-0408	Hardware/software inventory, file systems & metadata, hidden/encrypted files, ILookIX workflow, attribution
2(3)	"Project Monsoon Data Leak" – Seizure and imaging of a Dell Latitude 5420 laptop and Kingston USB drive from Research Lab R-204	REC-CDF-2026-0522	Physical acquisition, IXImager, hashing, ASB container, data carving, time issues, chain of custody
2(4)	"The Vanishing Quotations" – Recovery of deleted vendor quotation files from the Purchase Office workstation PO-WS03 using Volume Shadow Copies	REC-CDF-2026-0617	Evidence selection & classification, Event/Cloud/Lead analysis, VSC analysis, validation
2(5)	"Ghost Login" – Unauthorised use of student account 2419XX045 on the RECLearn online quiz portal, investigated across PC, browser, email, WhatsApp, mobile and cloud logs	REC-CDF-2026-0729	Windows/OS artefacts, browser, email/messaging, mobile, cloud evidence, correlation, challenges

General assumptions applied to all assignments: (i) the investigation is internal to the college and is authorised in writing by the Principal; (ii) a trained examiner from the college's Cyber Forensics Lab, assisted by a second examiner, performs the work; (iii) all analysis is performed on verified forensic copies; (iv) where police involvement becomes necessary (for example, an offence under the Information Technology Act, 2000), evidence is handed over with the full chain-of-custody record.

Assignment 2(1): Suspected Unauthorised Access to a College Server

Case Scenario: "Operation Midnight Paper" (Case REC-CDF-2026-0315)

On **Sunday, 15 March 2026 at 09:10**, Dr. A. Kavitha (fictitious), Examination Cell Coordinator of the CSE Department, reported that the folder **\\EXAMSRV-CSE02\Exams\Sem4_Apr2026\QP_Final** containing the final question papers for four theory courses had been accessed and copied on the night of **14 March 2026, between 23:00 and 23:30**. The server is a Windows Server 2019 file server; access is limited to three faculty accounts.

The server's Security log shows a successful network logon by the account **fac_rmohan** (a faculty member who was on leave in Madurai that weekend) from IP **172.16.3.117**, which the DHCP server maps to workstation **LAB3-WS17** in Programming Lab-3. A **SanDisk Cruzer Blade 32 GB USB drive** was found plugged into LAB3-WS17 on Monday morning by the lab assistant. A student volunteer, "Student S" (Reg. 2419XX0XX), had requested late-night lab access on 14 March for a hackathon.

Assumptions

- LAB3-WS17 runs Windows 11 Pro (NTFS) and was found switched OFF when the team arrived; the USB was still inserted.
- Server auditing (Event IDs 4624, 4634, 4663, 5140, 5145) was already enabled through Group Policy; logs had not rolled over.
- The lab CCTV and the hostel/lab gate register are available as non-digital corroborating sources.

Q1. Investigation phases for this case

The investigation follows the five-phase digital forensic model covered in the syllabus. Each phase produces documented output that becomes the input to the next phase, so that every conclusion can be traced back to an original, unaltered item of evidence.

Phase	Activities in "Operation Midnight Paper"	Output
1. Identification	Record the complaint; obtain written authorisation; define scope (who, what, when); identify all potential evidence sources – EXAMSRV-CSE02, LAB3-WS17, the USB drive, DHCP/firewall logs, CCTV, lab register; isolate the server share (change permissions, not shut down) to prevent further loss; photograph Lab-3 and the workstation with the USB in place.	Case file, scope note, evidence source list, scene photographs
2. Collection / Preservation	Label each item (E-01 to E-07); image the WS17 disk and the USB through a hardware write-blocker into E01 format; export server event logs (.evtx) and file-share audit logs with hash values; obtain DHCP lease log and firewall log; take MD5 and SHA-256 before and after acquisition; seal originals in tamper-evident bags; fill chain-of-custody forms.	Forensic images, hash sheet, chain-of-custody forms

Phase	Activities in “Operation Midnight Paper”	Output
3. Examination	Work only on verified copies. Parse NTFS ($MFT, $UsnJrnl, $LogFile); extract registry hives (SYSTEM, SOFTWARE, NTUSER.DAT) to read USBSTOR, MountedDevices and ShellBags; parse LNK files and Jump Lists; filter server logs for 4624/4663/5145 between 22:00 and 01:00; keyword search for course codes (e.g., “CS23431”) and file names of the question papers; hash files on the USB and compare with server originals.	Extracted artefacts, filtered log tables, hash-match list
4. Analysis	Correlate the server logon by fac_rmohan with the DHCP lease of 172.16.3.117 → LAB3-WS17; correlate USB serial (from USBSTOR) with the seized drive; build a super-timeline; determine which Windows profile on WS17 was active; check how fac_rmohan’s credentials could have been used (saved credentials, Credential Manager, shoulder-surfing, a password note); test alternative explanations (remote access, shared account, clock error).	Timeline, correlation matrix, hypotheses with evidential support
5. Presentation	Prepare a technical forensic report with hash values and methodology, plus a one-page executive summary in plain language for the Principal and Disciplinary Committee; present visual timeline; be prepared to explain the methodology and limitations if the matter goes to police or court.	Final report, executive summary, exhibits

Q2. Forensic investigation flowchart

The flowchart in Figure 1.1 shows the sequence of activities from the initial complaint to the presentation of findings. The decision points are the power state of each system (which decides whether volatile data is captured) and peer verification of findings before reporting.

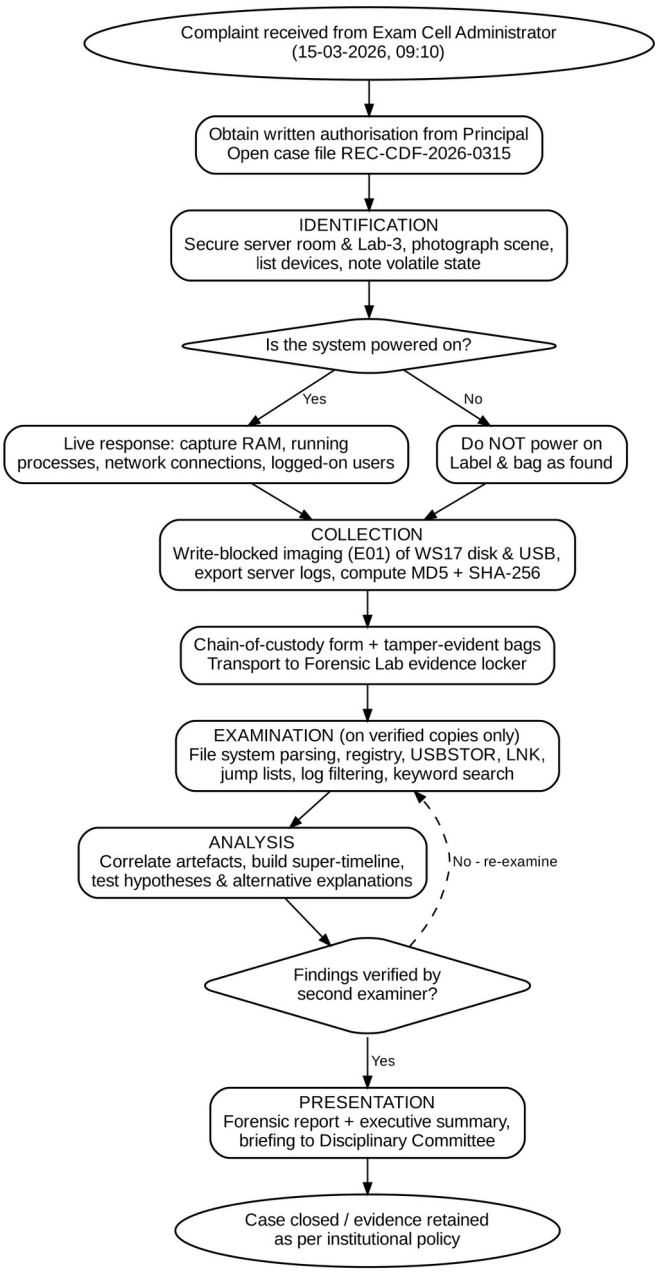


Figure 1.1: Forensic investigation flowchart for Case REC-CDF-2026-0315

Q3. Potential digital evidence and classification by investigative value

Seven items of potential evidence were identified. They are classified as **Primary** (directly shows the unauthorised act), **Corroborative** (supports or links primary evidence) or **Contextual** (gives background or helps exclude explanations).

ID	Evidence Item	What it may show	Volatility	Investigative Value
E-01	EXAMSRV-CSE02 Security & file-share audit logs (.evtx)	Logon by fac_rmohan (4624, Logon Type 3), share access (5140/5145), object access to QP files (4663), source IP 172.16.3.117	Medium (logs roll over)	Primary

ID	Evidence Item	What it may show	Volatility	Investigative Value
E-02	Disk of workstation LAB3-WS17 (forensic image)	User profile active at 23:00, mapped drive to \\EXAMSRV-CSE02, LNK/Jump Lists for QP files, USB connection records, Credential Manager entries	Low	Primary
E-03	SanDisk Cruzer Blade 32 GB USB (Serial 4C530001230923112345 – fictitious)	Copies of question papers; file creation times on the USB = time of copy; volume serial number	Low	Primary
E-04	Copied files (QP_CS23431.docx etc.) on the server and on the USB	Hash comparison proves the USB files are identical copies of the server originals	Low	Primary / Corroborative
E-05	DHCP lease log and firewall log	Maps IP 172.16.3.117 to MAC address and host name LAB3-WS17 at that time	High (short retention)	Corroborative
E-06	Administrator's written complaint and email	Defines the incident, affected files and time window	N/A	Contextual
E-07	Lab access register and CCTV footage of Lab-3 corridor	Who was physically present in Lab-3 at 23:00–23:30	Medium (CCTV overwrites in 7 days)	Corroborative (physical)

Q4. Precautions during collection and preservation

The evidence must be **reliable** (it truly represents what happened) and **defensible** (the institution can prove it was not altered). The following precautions apply:

- **Authorisation first:** collect only after written approval from the Principal, and only the devices within scope.
- **Do not power on LAB3-WS17:** booting Windows changes hundreds of files, timestamps and logs. Remove the disk in the lab and image it through a **hardware write-blocker**.
- **Do not browse the USB on any normal PC:** Windows would write to it (e.g., System Volume Information, thumbnail caches). Image it through a write-blocker.
- **Server stays online:** it is a production server, so perform a **logical, live collection** of the relevant logs and folder metadata using an approved tool; record every command, the operator and the time.
- **Hash everything:** compute MD5 and SHA-256 at acquisition; verify the hash again before every examination session.
- **Prioritise volatile/short-lived evidence:** DHCP logs and CCTV are overwritten quickly and must be preserved first (order of volatility).
- **Document the scene:** photographs, sketches, device labels, serial numbers, cable connections and the exact time on each device's clock against a reference clock.
- **Packaging and storage:** anti-static bags, tamper-evident seals signed across the seal, a locked evidence locker with an access register.

- **Chain of custody:** every hand-over (who, when, why, condition of seal) is recorded without gaps.
- **Minimise people:** the suspect account holder and the complainant should not handle the evidence.

Q5. Establishing a timeline of the suspected unauthorised access

Timestamps from all sources are first normalised to IST and adjusted for any clock drift measured during collection (here WS17's clock was found **2 minutes 10 seconds fast** against the NTP reference; the server was synchronised). Events are then merged into one super-timeline (for example using Plaso/log2timeline) and the key events are verified manually.

Time (IST)	Source	Event	Evidence ID
14-03-2026 22:41	Lab access register / CCTV	Student S signs in to Lab-3 for hackathon practice	E-07
22:47	WS17 Security log (4624, Type 2)	Local interactive logon to WS17 using generic lab account lab3user	E-02
23:04	WS17 registry USBSTOR / Event 6416 / setupapi.dev.log	SanDisk Cruzer Blade (serial …2345) first connected; assigned drive letter E:	E-02, E-03
23:06	Server Security log 4624 (Type 3)	Network logon by fac_rmohan from 172.16.3.117	E-01
23:06	DHCP lease log	172.16.3.117 leased to MAC 3C:52:82:xx:xx:17 = LAB3-WS17	E-05
23:07	Server 5140 / 5145	Share \\EXAMSRV-CSE02\Exams accessed; QP_Final folder enumerated	E-01
23:09 – 23:12	Server 4663 (ReadData)	Four QP files read by fac_rmohan	E-01
23:09 – 23:12	USB $MFT (created times)	Four files with identical names created on E:\ hack\ref\	E-03
23:12	Hash comparison	SHA-256 of USB files match server originals	E-04
23:14	WS17 LNK files / Jump Lists	QP_CS23431.docx opened from E:\hack\ref\	E-02
23:21	Server 4634	fac_rmohan network session logoff	E-01
23:28	Lab register / CCTV	Student S leaves Lab-3; USB left in WS17	E-07
15-03-2026 09:10	Complaint email	Incident reported by Exam Cell Coordinator	E-06

Table 1.2: Reconstructed timeline (normalised to IST, WS17 drift corrected)

The timeline shows that the copy occurred from LAB3-WS17 while the lab3user session was active, using fac_rmohan's credentials. The credential path is checked separately: the examiner found (hypothetically) a **Windows Credential Manager entry for EXAMSRV-CSE02** saved under lab3user on 02-03-2026, when Prof. R. Mohan had used WS17 for a lab demonstration, an important finding that explains how the faculty account was used without the faculty member being present.

Q6. Final forensic report and presentation to a non-technical authority

The report must allow another qualified examiner to repeat the work and reach the same result, while also allowing a non-technical committee to understand the outcome. The recommended structure is:

Section	Contents
1. Title page & case details	Case number, report date, examiner(s), requesting authority, classification "Confidential"
2. Executive summary	One page in plain language: what happened, when, from where, the confidence level, and recommended actions
3. Scope & authorisation	Questions asked by the authority; copy of authorisation; limitations of scope
4. Evidence list	Each exhibit with description, serial number, hash values, date received
5. Methodology & tools	Write-blocker used, imaging tool and version, analysis tools, hash verification steps
6. Findings	Numbered factual findings, each referencing the artefact, path, timestamp and exhibit
7. Timeline & analysis	Consolidated timeline, correlation, alternative explanations considered and why they were accepted/rejected
8. Conclusion	Opinion limited to what evidence supports (e.g., "the files were copied from LAB3-WS17 to exhibit E-03 using account fac_rmohan"); no assumptions about intent
9. Recommendations	Disable saved credentials on lab PCs, enforce MFA for exam server, separate exam network, review audit retention
10. Appendices	Hash sheets, chain-of-custody forms, log extracts, screenshots, glossary
11. Declaration & signature	Examiner's statement of truth, signature, date

Presenting to non-technical authorities: use plain terms ("a record in the server's security diary" instead of "Event ID 4624, Logon Type 3"); show a simple visual timeline; use analogies (a hash value is a "digital fingerprint" of a file); state clearly what is **fact**, what is **inference** and what is **unknown**; avoid naming a person as guilty, the report links activity to accounts, devices and times, and the committee decides responsibility; keep technical detail in appendices for verification.

Q7. Legal and ethical considerations in a college environment

- **Authorisation and policy:** collection must be backed by the college IT Acceptable Use Policy, which students and staff accept, and by written authorisation from the competent authority.
- **Indian legal framework:** unauthorised access and copying are covered under **Sections 43 and 66 of the Information Technology Act, 2000**; electronic evidence admissibility in court requires a certificate under **Section 63 of the Bharatiya Sakshya Adhiniyam, 2023** (formerly Section 65B of the Indian Evidence Act). Personal data processing should respect the **Digital Personal Data Protection Act, 2023**.
- **Privacy and proportionality:** examine only data relevant to the case; personal photos, chats or unrelated documents of students and staff must not be viewed, copied or disclosed.

- **Presumption of innocence:** the account owner fac_rmohan and Student S must both be treated as persons of interest, not culprits, until evidence is analysed; their right to be heard must be respected.
- **Confidentiality:** findings are shared only with the authorised committee; question papers already exposed must be replaced (a remedial action, not a forensic one).
- **Minors / students:** the involvement of parents or guardians and the college grievance procedure should be followed where required.
- **Examiner integrity:** neutrality, competence, accurate reporting of limitations and no alteration of evidence.

Assignment 2(2): Linking a Suspicious File to a User Using File-System Evidence

Case Scenario: "The GradeSync Case" (Case REC-CDF-2026-0408)

The Training & Placement Office uses a shared desktop **PC-TPO-04** (HP ProDesk 400 G7) with four Windows accounts: **tpo_admin**, **staff_ravi**, **staff_meena** and **guest**. On **8 April 2026** the antivirus quarantined **GradeSync.exe**, found in a hidden folder **C:\Users\Public\Libraries\.cache**, after it attempted to upload files to an external IP. Investigators also found three password-protected archives (**placement_list_2026.7z**, **offers_final.7z**, **cgpa_master.7z**), several files with the Hidden + System attributes, and browser artefacts in two profiles.

The Placement Officer suspects that the account **staff_ravi** created and ran GradeSync.exe to exfiltrate student CGPA and offer data. The task is to examine the environment and use **ILookIX** to determine which account, if any, can be linked to the file.

Assumptions

- Windows 10 Pro 22H2, single 512 GB NVMe SSD, NTFS, BitLocker not enabled; the PC is part of no domain (local accounts).
- A verified E01 image (SHA-256 recorded) was acquired on 08-04-2026 after RAM capture; analysis is on a copy.

Q8. Hardware and software components to document before analysis

Documenting the environment establishes the baseline against which artefacts are interpreted (for example, the time zone affects every timestamp, and the file system determines which metadata exist).

Category	Component	Recorded Detail (hypothetical)	Why it matters
Hardware	System make/model/serial	HP ProDesk 400 G7, S/N 8CC0XXXX4	Identifies the exact exhibit
Hardware	Storage	Samsung PM991 512 GB NVMe, S/N S4XXNX0XXXX	Determines imaging method; SSD TRIM affects deleted-file recovery
Hardware	RAM / CPU	8 GB DDR4 / Intel Core i5-10500	Size of memory capture
Hardware	Peripherals & ports	2 × USB 3.0, 4 × USB 2.0, network card MAC 70:5A:0F:xx:xx:04	Links network logs to the PC
Hardware	BIOS/UEFI clock	BIOS time 2 min 40 s slow vs. NTP	Clock-offset correction
Software	Operating system	Windows 10 Pro 22H2, build 19045, install date 12-07-2023	Artefact locations & formats
Software	File system	NTFS, cluster 4 KB, partitions: EFI, MSR, C: (476 GB), Recovery	$MFT, ADS, $UsnJrnl availability
Software	Time zone setting	(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi	Converts UTC timestamps
Software	User accounts & SIDs	tpo_admin (RID 1001, admin), staff_ravi (1002), staff_meena (1003), guest (1004)	Mapping artefacts to accounts

Category	Component	Recorded Detail (hypothetical)	Why it matters
Software	Installed applications	Chrome 123, Edge, 7-Zip 23.01, MS Office 2019, Quick Heal AV	Explains encrypted .7z files & browser artefacts
Software	Security settings	UAC on, audit logon events enabled, no full-disk encryption, shared password sticky-note found	Credential sharing hypothesis

Table 2.1: System inventory of PC-TPO-04

Q9. How storage devices, OS, file systems and metadata help locate evidence

- **Storage devices:** the partition table (GPT) reveals all volumes, including hidden recovery partitions and unallocated space where deleted data or hidden volumes may exist. On SSDs, TRIM can wipe deleted clusters, so the examiner notes this limitation early.
- **Operating system:** Windows keeps per-user artefacts in each profile (C:\Users\<name>\) and in per-user registry hives (NTUSER.DAT, UsrClass.dat). Execution artefacts such as **Prefetch** (GRADESYNC.EXE-XXXXXXXX.pf), **Amcache.hve**, **ShimCache**, **UserAssist** and **BAM** (Background Activity Moderator, keyed by user SID) show whether and by whom a program was run.
- **File system (NTFS):** the **$MFT** stores each file's record, including the **$STANDARD_INFORMATION** and **$FILE_NAME** timestamps (MACB – Modified, Accessed, MFT-Changed, Born), the parent folder, and the **owner SID** in the security descriptor. **$UsnJrnl:$J** records every create/rename/delete operation; **$LogFile** holds recent transactions; **Alternate Data Streams** such as **Zone.Identifier** show whether a file was downloaded from the Internet and from which URL.
- **Metadata:** application metadata (Office "Author", "Last saved by"), EXIF data, PE header compile time of the executable, and 7-Zip archive headers (file names inside are visible even when encrypted unless header encryption is used) help link files to people and times.

In this case, the owner SID of GradeSync.exe, the Zone.Identifier stream pointing to a download URL, the BAM entry under one SID and the browser history of that same profile together locate and link the evidence.

Q10. Effect of hidden and encrypted files on the investigation

Issue	Effect on Investigation	Evidence-handling consideration
Hidden / System attribute files	Not visible in Explorer; may be overlooked in a casual review; often used by malware (C:\Users\Public\ Libraries\.cache\)	Forensic tools show all files regardless of attributes; record the attribute flags as evidence of concealment intent
Alternate Data Streams	Data can be hidden inside another file's ADS	Use tools that enumerate ADS; hash both main and alternate streams
File-extension mismatch	A file renamed (e.g., report.jpg actually a ZIP) evades filters	Run signature (magic number) analysis
Password-protected .7z archives	Content cannot be examined without the password; the brute-force may take very long	Do not attempt guesses on original; search memory image, browser saved passwords, notes and sticky-notes for passwords; legally request the password from the account holder; document the encryption algorithm (AES-256)

Issue	Effect on Investigation	Evidence-handling consideration
Encrypted volumes / BitLocker / VeraCrypt	Whole data may be inaccessible after shutdown	Capture RAM first if the system is live; seek recovery key from Microsoft account/IT admin
Anti-forensic wiping	Deleted hidden files may be overwritten	Look for traces in $UsnJrnl, VSCs, LNK files and Prefetch which survive deletion

Encryption itself is not proof of wrongdoing, staff may legitimately protect confidential placement lists. The report must therefore state only that the files were encrypted, with what tool, when and by which account.

Q11. Evidence-linkage table

Each artefact type is correlated against the four accounts. The pattern of independent artefacts pointing to the same SID strengthens attribution.

Artefact	Location / Path	Timestamp (IST)	Account / SID linked	Linkage Strength
File owner SID of GradeSync.exe	$MFT record of C:\Users\Public\Libraries\.cache\GradeSync.exe	Born 02-04-2026 16:52:10	S-1-5-21-…-1002 (staff_ravi)	Strong
Zone.Identifier ADS	GradeSync.exe:Zone.Identifier → HostUrl=hxxps://filedrop[.]example/gs.exe	02-04-2026 16:52	–	Supporting
Chrome download record	C:\Users\staff_ravi\AppData\Local\Google\Chrome\User Data\Default\History (downloads table)	02-04-2026 16:52:08	staff_ravi	Strong
Browser search history	Same History file (urls table): “how to sync excel data to remote server”	02-04-2026 16:40	staff_ravi	Supporting
BAM execution entry	SYSTEM\…\bam\State\UserSettings\S-1-5-21-…-1002	Last run 07-04-2026 18:31	staff_ravi	Strong
Prefetch file	C:\Windows\Prefetch\GRADESYNC.EXE-1A2B3C4D.pf (run count 5)	Runs on 02, 03, 05, 07 Apr	System-wide (time-matched to staff_ravi logons)	Supporting
Logon events	Security.evtx 4624 Type 2	staff_ravi logged on 16:35–17:20 on 02-04	staff_ravi	Corroborating
LNK file	C:\Users\staff_ravi\AppData\Roaming\Microsoft\Windows\Recent\cgpa_master.7z.lnk	05-04-2026 11:15	staff_ravi	Strong (file access)
7-Zip archive metadata	cgpa_master.7z headers: contains CGPA_2026_all.xlsx; created 05-04 11:14	05-04-2026 11:14	Owner SID … 1003 (staff_meena)	Conflicting – needs explanation
Office metadata	CGPA_2026_all.xlsx (in Documents of tpo_admin) – Last saved by “Meena S”	04-04-2026 15:02	staff_meena	Contextual
Antivirus log	Quick Heal quarantine log – blocked outbound connection	08-04-2026 10:03	Process running under staff_ravi	Strong

Table 2.2: Evidence-linkage matrix for GradeSync.exe

Q12. ILookIX import and analysis workflow

ILookIX (Perlustro) is a forensic analysis suite that ingests images (E01, raw/dd, its own ASB/IXImager formats), parses file systems and presents artefacts with tagging, search and timeline capabilities. The intended workflow is shown in Figure 2.1 and explained below.

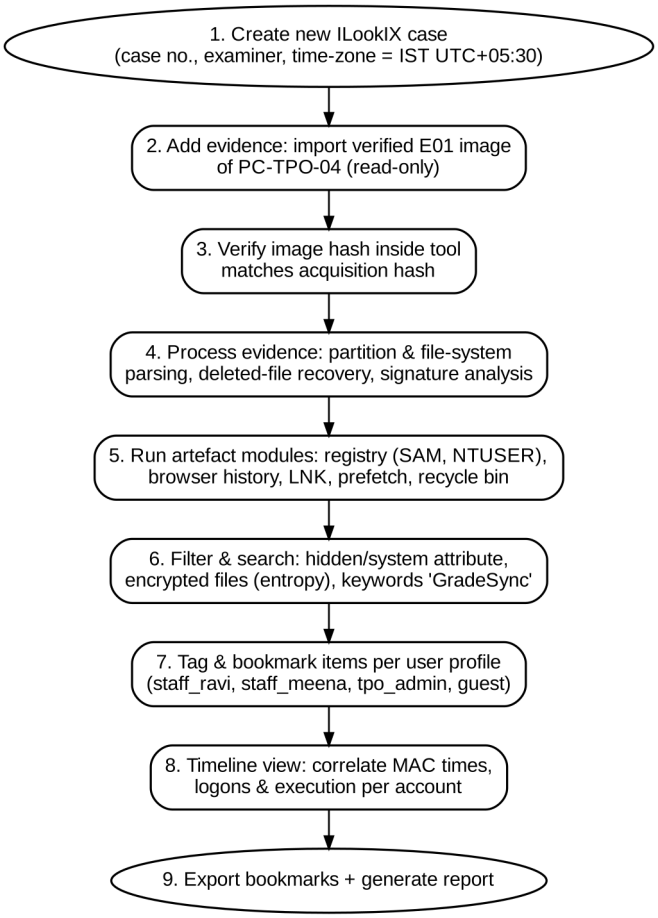


Figure 2.1: ILookIX analysis workflow for PC-TPO-04

1. **Case creation:** create a case “REC-CDF-2026-0408”, enter examiner details and set the display time zone to IST so that UTC timestamps are converted consistently.
2. **Evidence import:** add the verified image of PC-TPO-04 as read-only evidence. ILookIX never writes to the source image.
3. **Integrity verification:** let ILookIX compute the image hash and compare it with the acquisition hash recorded on the chain-of-custody form; record the result.
4. **Processing:** allow the tool to identify partitions, parse NTFS (including $MFT, deleted entries, ADS), recover deleted files, run file-signature analysis and calculate hashes for known-file filtering (to eliminate known Windows files and flag known-bad hashes).
5. **Artefact extraction:** use the registry viewer for SAM (accounts and SIDs), NTUSER.DAT (UserAssist, RecentDocs), SYSTEM (BAM, USB); view browser history, LNK files, Prefetch and Recycle Bin per user.
6. **Filtering and searching:** filter on Hidden/System attributes, high-entropy files (indicating encryption), the name “GradeSync”, the download domain and the archive names; use indexed keyword search.

7. **Tagging:** bookmark relevant items into categories such as “Execution”, “Download”, “Access to encrypted archives”, separated by user account.
8. **Timeline:** generate a timeline of the bookmarked items, compare with logon/logoff sessions, and identify overlaps and gaps.
9. **Reporting:** export bookmarked files and create a report containing paths, timestamps, hash values and examiner notes. Validate key findings with a second tool (e.g., Autopsy or Registry Explorer).

Q13. Direct-linkage artefacts versus supporting/contextual artefacts

Aspect	Direct-linkage artefact	Supporting / contextual artefact
Definition	Records a specific action by a specific account/SID at a specific time	Shows circumstances, possibility or general activity, but not who performed the specific act
Examples in this case	BAM entry under SID …1002 for GradeSync.exe; Chrome download entry in staff_ravi’s profile; LNK file in staff_ravi’s Recent folder	Prefetch run count (system-wide, not per-user); Zone.Identifier URL; Office “Last saved by” name (user-editable); sticky-note with shared password
Weight	High, but still subject to the possibility that someone else used that account	Low to medium individually; strengthens the case when several agree
Use in report	Stated as finding: “Account staff_ravi executed …”	Stated as corroboration or context: “consistent with …”

Q14. Limitations and alternative explanations before attribution

1. **Account is not person:** a password sticky-note was found on the monitor; any staff member could have logged in as staff_ravi. Logon times must be compared with attendance, CCTV and biometric records.
2. **Malware or remote execution:** GradeSync.exe could have been dropped by another malicious program or run via a scheduled task, creating BAM/Prefetch entries without the user knowingly running it. Check Task Scheduler, Run keys and parent-process data.
3. **Timestamp manipulation:** $STANDARD_INFORMATION times can be altered (“timestomping”); compare with $FILE_NAME times and $UsnJrnl.
4. **Conflicting ownership of cgpa_master.7z:** the archive’s owner SID is staff_meena; she may have created it legitimately for the Placement Officer and staff_ravi only opened it. Both roles must be reported fairly.
5. **SSD TRIM and missing data:** deleted evidence may be permanently lost, so absence of evidence is not evidence of absence.
6. **Clock offset:** the BIOS clock was 2 min 40 s slow; uncorrected times could mis-align with external logs.

Conclusion

Multiple independent artefacts (owner SID, Chrome download record, BAM entry, LNK file and the antivirus log) consistently link the **download, execution and use of GradeSync.exe to the Windows account staff_ravi** between 02-04-2026 and 08-04-2026. However, because the account password was shared on a sticky-note and one archive was created by staff_meena, the

report concludes that the activity is attributable to the **account**; attribution to a **person** requires corroboration from non-digital sources (attendance, CCTV, interview).

Assignment 2(3): Forensic Imaging and Safekeeping of a Suspect Storage Device

Case Scenario: "Project Monsoon Data Leak" (Case REC-CDF-2026-0522)

Research Lab R-204 runs a funded project ("Monsoon") on rainfall-prediction models. On **22 May 2026**, the Principal Investigator found the project's unpublished dataset and trained model posted on a public code-sharing site. A project intern was assigned a **Dell Latitude 5420 laptop** (Service Tag 7XKXXX3, 512 GB SK hynix NVMe SSD, Windows 11) with an attached **Kingston DataTraveler Exodia 64 GB USB drive**. The laptop and USB were seized at **14:30 on 22 May 2026** for physical acquisition using **IXImager**, storage in the **ASB** container format and later analysis.

Assumptions

- The laptop was powered on and unlocked when found; BitLocker was enabled on C:.
- The college Cyber Forensics Lab has a Tableau T35u / T8u hardware write-blocker, a forensic workstation with IXImager and ILookIX, and an evidence locker.

Q15. Step-by-step procedure for collecting and transporting the laptop and USB

1. **Verify authority:** confirm written seizure authorisation and identify the lead examiner and witness.
2. **Secure the scene:** stop all persons from touching the laptop; note who was present.
3. **Photograph and document:** take photos of the screen, open applications, USB position, cables and labels; record make, model, service tag and USB serial; note system time vs. reference clock (laptop showed 14:31:20 when reference clock showed 14:30:05 → +75 s).
4. **Volatile data capture (laptop ON):** because BitLocker is enabled, shutting down would lock the data. Capture RAM using a trusted tool from the examiner's USB (e.g., Magnet RAM Capture/FTK Imager Lite), and record the BitLocker recovery key (manage-bde -protectors -get C:) and running processes/network connections. Every action is written in the field log with time.
5. **Remove the USB:** only after RAM capture, use "safely remove" to avoid file-system corruption, or pull it if malicious activity is in progress; record the time.
6. **Power down:** perform a hard power-off (hold power button / remove battery) to avoid shutdown scripts that might wipe data.
7. **Label:** E-01 laptop, E-02 USB, E-03 charger, E-04 RAM image (on examiner's sterile drive).
8. **Package:** place in anti-static bags, then in tamper-evident evidence bags; sign across seals; place in a padded box. Keep away from magnets, heat and moisture.
9. **Chain of custody:** complete the form at the scene; the witness co-signs.
10. **Transport:** carry personally to the lab in a locked case, without leaving it unattended; on arrival, log into the evidence register and store in the locker until imaging.

Q16. Purpose of physical acquisition and why analysis uses a forensic image

Physical acquisition creates a bit-for-bit copy of the entire storage medium – every sector, including allocated files, unallocated space, slack space, partition gaps and hidden areas

(HPA/DCO where applicable). This differs from a **logical acquisition**, which copies only active files visible through the file system. Physical acquisition is needed here because the intern may have deleted the dataset after uploading it; remnants may exist only in unallocated space or slack.

Analysis is performed on a **forensic image** rather than the original because:

- Any access to the original risks changing it (timestamps, logs, wear-levelling on SSD), which would damage admissibility.
- The image can be copied multiple times so that several examiners/tools can work in parallel, and a mistake on a copy does not destroy evidence.
- The original is sealed and preserved for re-verification by the defence or a court-appointed expert.
- Hash verification proves that the image is identical to the original, so findings on the image are equally valid for the original.

Q17. Forensic imaging workflow using IXImager

IXImager is Perlustro's forensic imaging tool, designed to work with ILookIX. It runs from a bootable environment or the forensic workstation, supports write-protected acquisition and stores output in its compressed, encrypted **ASB** format. Figure 3.1 shows the overall flow.

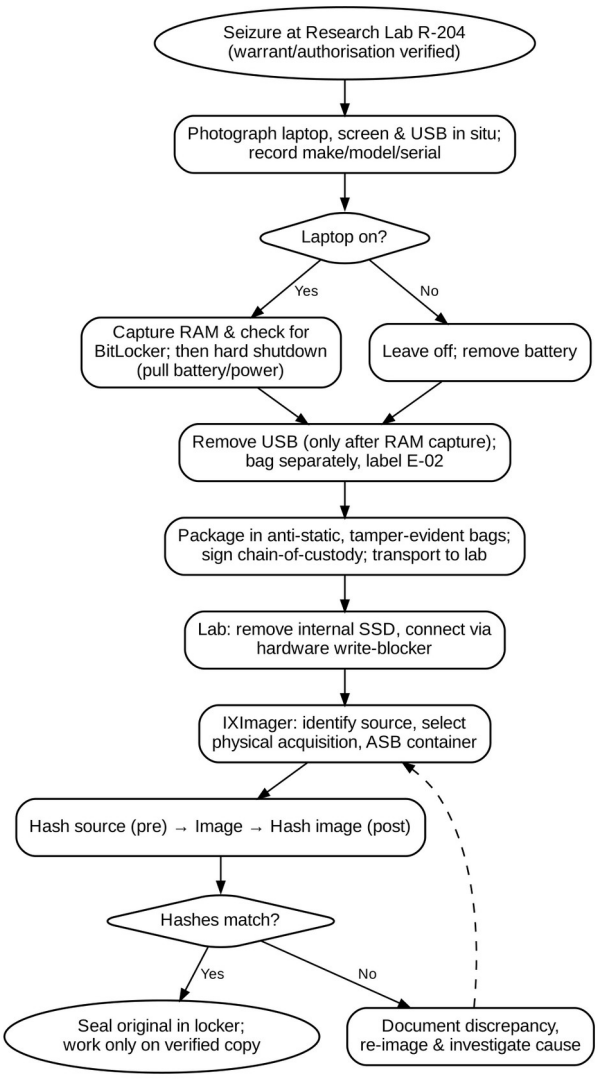


Figure 3.1: Seizure and acquisition flowchart using IXImager

Step	Action	Documentation
1. Source identification	Remove the NVMe SSD from the laptop (or boot IXImager on the laptop if removal is impossible). Record SSD model, serial, capacity (512,110,190,592 bytes) and the USB details.	Photo of drive label; device details sheet
2. Write protection	Connect SSD via hardware write-blocker (Tableau T35u with NVMe adapter); confirm “Write Blocked” LED; test that the OS shows the device read-only. For the USB, use a USB write-blocker.	Write-blocker model, serial, firmware, test result
3. Destination preparation	Use a forensically wiped, larger target drive (2 TB); record its serial and wiping log.	Target drive wipe certificate
4. Configure IXImager	Select source physical device (not a partition) → physical acquisition; output format ASB; enable compression and AES encryption with case password; enter case number, evidence number, examiner, description; set segment size.	Screenshot of settings
5. Pre-acquisition hash	Hash the source (MD5 + SHA-256) – IXImager calculates while reading.	Hash values recorded

Step	Action	Documentation
6. Imaging	Start acquisition; note start/end time; log any read errors or bad sectors (and how they are filled).	Acquisition log
7. Verification	Run verification: re-read the image and compute the hash; compare with the source hash.	Verification report
8. Second copy	Make a working copy from the verified master image; verify its hash.	Copy log
9. Seal & store	Reassemble and reseal the original; return to locker; update chain of custody.	CoC entry

Q18. Using hash values to demonstrate integrity

A cryptographic hash function (MD5, SHA-1, SHA-256) produces a fixed-length value from data; changing even one bit of the input changes the hash completely (avalanche effect), and it is computationally infeasible to find two different inputs with the same SHA-256 hash. Hence the hash acts as a **digital fingerprint**.

Point of Check	Hash Computed On	Example (SHA-256, truncated)	Interpretation
At acquisition (pre)	Source SSD via write-blocker	9f4c…a21e	Baseline fingerprint of the original
At acquisition (post)	ASB image file	9f4c…a21e	Match → image is an exact copy
Working copy created	Working copy	9f4c…a21e	Match → working copy is valid
Before each analysis session	Working copy	9f4c…a21e	Match → no change since last session
Court / re-verification	Original SSD (re-hashed)	9f4c…a21e	Match → original unaltered while in custody

Two algorithms (MD5 and SHA-256) are recorded so that weaknesses in one (MD5 collisions) do not undermine integrity. A mismatch must be documented and explained (e.g., a failing drive with unstable sectors), never ignored. For SSDs, hashing the source a second time may differ due to firmware behaviour on unreadable/ trimmed areas; this is noted as a known limitation.

Q19. Role of the ASB container and accompanying information

An **ASB container** is the evidence file format produced by IXImager and read by ILookIX. Like the EnCase E01 or AFF4 formats, it is a **forensic container**: it stores the bit-stream image together with case metadata and integrity data in one managed file set. Its roles are:

- **Storage:** holds the complete sector-by-sector image, compressed to save space and split into segments for easier handling.
- **Protection:** supports encryption so that sensitive seized data (here, unpublished research) is protected during storage and transfer.
- **Integrity:** embeds hash values/checksums so the tool can verify that the container has not been altered or corrupted.

- **Management:** carries case information and acquisition logs, so the evidence is self-describing when imported into ILookIX.

Information that should accompany the evidence container:

Field	Example Entry
Case number / evidence number	REC-CDF-2026-0522 / E-01-IMG
Examiner name & witness	Examiner: Lab Forensic Officer; Witness: HoD nominee
Source device details	SK hynix BC711 512 GB NVMe, S/N CY0XXXXXXX, from Dell Latitude 5420 ST 7XKXXX3
Acquisition type & tool	Physical; IXImager (version noted), write-blocker Tableau T35u
Date/time & time zone	Start 23-05-2026 10:12 IST, end 11:48 IST; device clock +75 s
Hash values	MD5 and SHA-256 of source and image; verification result
Errors / notes	No bad sectors; BitLocker recovery key stored separately in sealed envelope
Storage location	Evidence server share \\FORLAB\Cases\0522 and offline backup drive

Q20. Data carving and when it is necessary

Data carving is the recovery of files from raw data based on file **content** rather than file-system metadata. The carver scans unallocated space (or the whole image) for known **headers and footers** (magic numbers) and extracts the bytes between them. Examples: JPEG (FF D8 FF … FF D9), PDF (%PDF … %%EOF), ZIP/DOCX/XLSX (50 4B 03 04), CSV/text by keyword patterns.

Carving is necessary when:

- Files were deleted and their MFT entries were reused or overwritten, so the file system no longer points to the data.
- The USB drive was **quick-formatted**, destroying the FAT/exFAT directory structure, likely in this case, as the Kingston drive shows a new volume created on 21-05-2026.
- The partition table or file system is corrupted or deliberately damaged.
- Data resides in slack space, page file (pagefile.sys), hibernation file, or memory dumps.

Limitations: carved files lose names, paths and timestamps; fragmented files may be recovered incompletely; false positives are common. In this case the examiner would carve the USB image for ZIP/CSV/H5 (model) signatures and compare hashes of carved files with the leaked dataset.

Q21. Date/time problems and their documentation

Problem	Description	How to document / handle
Time zones	NTFS stores UTC; FAT/exFAT on USB stores local time; logs may be in either	Record the system time-zone setting (registry TimeZoneInformation); state the zone of every reported time
Clock drift	Device clock differs from real time	Compare device clock with a trusted reference at seizure (+75 s here); apply and document the correction

Problem	Description	How to document / handle
Daylight saving	Not in India, but cloud servers (e.g., code-sharing site) may use US/EU time with DST	Convert to UTC first, then to IST
Timestamp resolution	NTFS 100 ns, FAT 2 s for modification, exFAT 10 ms	Do not over-interpret small differences
Copy/move behaviour	Copying a file creates a new “Created” time but keeps the “Modified” time, so Created > Modified is normal for copies	Explain this in the report to avoid false conclusions
Deliberate manipulation	Timestomping or changing system clock	Compare $SI vs $FN times, $UsnJrnl, event log 4616 (system time changed)

Q22. Chain-of-custody record for the seized laptop

#	Date & Time (IST)	Released by	Received by	Purpose / Action	Seal No. & Condition
1	22-05-2026 14:30	– (seized at R-204)	Lead Examiner	Seizure, photography, RAM capture, power-off	TE-2205-01 applied
2	22-05-2026 15:20	Lead Examiner	Evidence Custodian, Forensic Lab	Transport and storage in locker L-3	TE-2205-01 intact
3	23-05-2026 09:55	Evidence Custodian	Lead Examiner	Removal of SSD, imaging with IXImager	Seal broken in witness presence; new seal TE-2305-02
4	23-05-2026 12:10	Lead Examiner	Evidence Custodian	Return of original after verified imaging	TE-2305-02 intact
5	25-05-2026 10:00	Evidence Custodian	Second Examiner	Working copy of ASB image (original not removed)	Image hash verified
6	30-05-2026 16:00	Second Examiner	Evidence Custodian	Working copy returned; analysis complete	Hash verified

Table 3.4: Chain-of-custody form – Exhibit E-01 (Dell Latitude 5420, ST 7XKXXX3)

A similar form is maintained separately for E-02 (Kingston USB) and E-04 (RAM image). Each entry carries the signatures of both releasing and receiving persons.

Assignment 2(4): Recovery of Deleted Evidence from Volume Shadow Copies

Case Scenario: "The Vanishing Quotations" (Case REC-CDF-2026-0617)

The college Purchase Office invited quotations for 120 laboratory computers. An internal audit on **17 June 2026** found that two genuine vendor quotations (**Quote_VendorA_Rs62L.pdf** and **Quote_VendorB_Rs58L.pdf**) were missing from the shared tender folder and that a higher-priced quotation from a third vendor had been approved. Clerk "Employee P" is suspected of having deleted the files, as well as Chrome history showing communication with the third vendor, from workstation **PO-WS03** (Windows 10, NTFS).

A forensic image of PO-WS03 is available. **Volume Shadow Copies (VSCs)** created on 05-06-2026, 10-06-2026 and 14-06-2026 are present. The task is to select relevant evidence, recover deleted artefacts and reconstruct the events.

Assumptions

- System Protection was enabled on C: with restore points created by Windows Update and a weekly scheduled task.
- The tender folder was synchronised to the office OneDrive for Business account, whose audit logs are available from the college IT admin.

Q23. Selecting and categorising evidence before detailed analysis

A forensic image of a 1 TB office PC contains hundreds of thousands of files; most are irrelevant. Selection (triage) focuses the examination, reduces time and minimises privacy intrusion.

1. **Define investigative questions:** Were the quotations present? When and by whom were they deleted? Was browser history cleared? Was there contact with the third vendor?
2. **Define the relevant time window:** quotations received 03-06-2026; approval made 15-06-2026; window 01-06 to 17-06-2026.
3. **Known-file filtering:** remove known OS/application files using hash sets (e.g., NSRL); this removes irrelevant data without deleting it from the image.
4. **Keyword and pattern filtering:** "Quote", vendor names, "tender", "Rs62L", email domains of vendors.
5. **Category assignment:** tag remaining items into the classes defined in Q24.
6. **Prioritisation:** examine primary categories first (deleted/recovered and suspicious), then supporting context.
7. **Document exclusions:** record what was excluded and why, so the selection itself can be reviewed.

Q24. Evidence classification scheme

Class	Definition	Examples in PO-WS03	Priority
A. User-created files	Documents created or edited by a user	Comparative_Statement_Lab_PCs.xlsx , approval note .docx, scanned quotations	High

Class	Definition	Examples in PO-WS03	Priority
B. System / application files	Files created by OS or applications; often logs and artefacts	$MFT, $UsnJrnl, Event logs, Prefetch, registry hives, OneDrive logs (SyncDiagnostics.log)	Medium (artefact source)
C. Potentially suspicious files	Files indicating anti-forensic or unusual activity	CCleaner.exe prefetch, browser-cleaning extension, renamed file “tmp_0614.dat” (actually PDF)	High
D. Deleted / recovered artefacts	Items no longer active but recovered from Recycle Bin, unallocated space or VSCs	$I/$R files in $Recycle.Bin, quotations recovered from VSC-2 & VSC-3, older Chrome History DB	Very High
E. Supporting / contextual evidence	Information that explains circumstances but does not prove the act	Logon/logoff events, OneDrive audit log, email from Vendor C, office attendance register	Medium

Table 4.1: Evidence classification table

Q25. How Event Analysis assists in reconstructing activities

Event Analysis examines time-stamped records generated by the OS and applications and arranges them into an ordered sequence of actions. Sources include Windows event logs (Security 4624/4634 logon/logoff, 4663 object access, 4660 object deleted, System 7045 service install), $UsnJrnl (FILE_DELETE, RENAME_NEW_NAME reason codes), Prefetch (program execution), Jump Lists and LNK files (file opens), and browser databases. By correlating these events the examiner can answer “what happened, in what order, by which account”, e.g., Employee P logged on at 17:40, opened the tender folder, ran CCleaner at 18:02 and the files show a USN delete record at 18:05. Event Analysis also exposes **gaps** (cleared logs: Event ID 1102) that are themselves suspicious.

Q26. Purpose of Cloud Analysis and Lead Analysis

	Cloud Analysis	Lead Analysis
Purpose	Identifies and examines evidence held in cloud services linked to the device – sync clients, cached tokens, local sync folders and provider audit logs	Generates and follows investigative leads – new persons, accounts, devices, keywords or locations discovered during examination
Sources in this case	OneDrive for Business sync folder, OneDrive logs, Microsoft 365 unified audit log (FileDeleted, FileDownloaded), Gmail web cache	Email from Vendor C, phone number in a sticky note, USB device in registry, contacts in Outlook
Questions it helps answer	Was the deletion synchronised to the cloud? Does the cloud recycle bin still hold the files? From which IP/device was the delete performed? Were the files shared externally before deletion?	Who else communicated with Vendor C? Was a USB used to take copies? Is there another device (personal phone/laptop) to be seized? Which leads justify further authorisation?

Q27. How Volume Shadow Copy analysis helps

The **Volume Shadow Copy Service (VSS)** creates point-in-time snapshots of an NTFS volume, mainly for System Restore and backups. VSS uses a **copy-on-write** mechanism at block level: when a block of the live volume is about to change, the old block is first copied into

the shadow storage area (in System Volume Information). Each shadow copy therefore preserves how the volume looked at its creation time.

- **Recovery of deleted files:** a file deleted on 14-06 may still exist completely in a snapshot taken on 10-06.
- **Comparison of versions:** the comparative-statement spreadsheet can be compared across snapshots to show when Vendor A and B's figures were removed.
- **Recovery of older artefacts:** earlier copies of registry hives, event logs (before being cleared) and the Chrome History database (before history was cleared).
- **Detecting anti-forensics:** deletion of shadow copies (vssadmin delete shadows) itself leaves traces and indicates intent.

Figure 4.1 illustrates the recovery workflow.

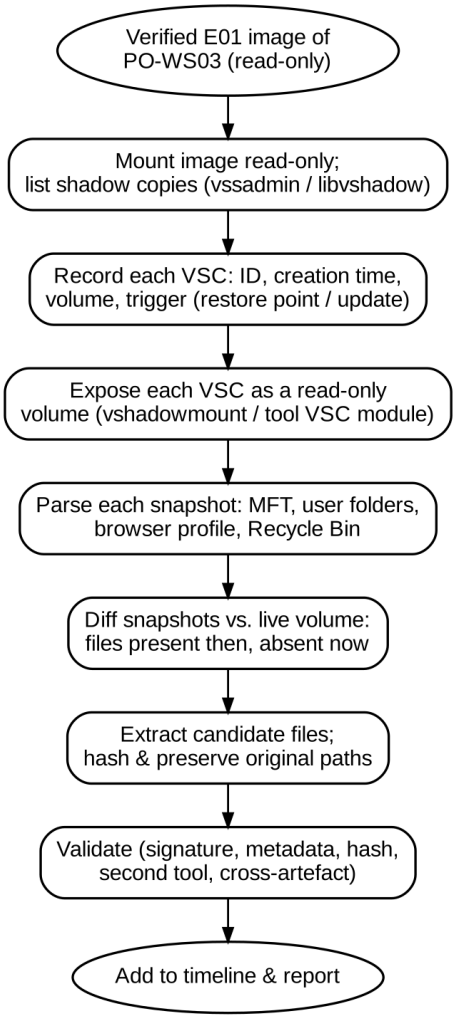


Figure 4.1: Volume Shadow Copy recovery workflow

VSC ID	Created (IST)	Trigger	Quotation files present?	Chrome History
VSC-1 {3a9…}	05-06-2026 11:00	Windows Update KB50xxxx	Vendor A only (B not yet received)	Contains vendor-C domain visits
VSC-2 {7c1…}	10-06-2026 02:00	Weekly scheduled restore point	Vendor A & B present	Contains vendor-C visits & webmail

VSC ID	Created (IST)	Trigger	Quotation files present?	Chrome History
VSC-3 {b42…}	14-06-2026 13:30	Software install (Adobe Reader update)	Vendor A & B present	Present
Live volume	17-06-2026 (image)	–	Both missing	History cleared 14-06 18:03

Table 4.2: Shadow copies identified on PO-WS03

Q28. Hypothetical timeline: create → modify → copy → delete → recover

Date & Time (IST)	Action	Artefact / Source	Class
03-06-2026 10:22	Quote_VendorA_Rs62L.pdf created (saved from email attachment) in D:\ Tender\LabPC2026\	$MFT Born time; Outlook attachment cache	A
05-06-2026 11:00	VSC-1 created (contains Vendor A file)	VSS catalog	B
08-06-2026 15:40	Quote_VendorB_Rs58L.pdf created	$MFT; OneDrive upload log	A
09-06-2026 12:15	Comparative_Statement.xlsx modified – both vendors listed	$SI Modified; Office metadata	A
10-06-2026 02:00	VSC-2 created (both files present)	VSS catalog	B
12-06-2026 17:51	Both quotations copied to USB “HP v236w” (S/N 2A1…); new Created times on USB copies	USBSTOR; LNK to E:\ Quote_VendorA…; Jump List	C/E
14-06-2026 13:30	VSC-3 created	VSS catalog	B
14-06-2026 17:58	Comparative_Statement.xlsx modified – Vendors A & B rows removed	$UsnJrnl DATA_OVERWRITE; compare with VSC-3 version	A/D
14-06-2026 18:02	CCleaner executed	Prefetch CCLEANER64.EXE-xxxx.pf	C
14-06-2026 18:03	Chrome history cleared	History DB empty for range; CCleaner log; VSC-3 copy has history	C/D
14-06-2026 18:05	Both quotation PDFs deleted (Shift+Delete, bypassing Recycle Bin)	$UsnJrnl FILE_DELETE CLOSE; OneDrive audit FileDeleted by user P	D/E
17-06-2026 11:30	Forensic image acquired	Acquisition log	–
19-06-2026 15:10	Files recovered from VSC-3; SHA-256 matches copies in vendor emails	Recovered files + hash report	D

Table 4.3: Event timeline for “The Vanishing Quotations”

Q29. Validating the recovered artefact before reporting

Recovered data must be shown to be authentic, complete and correctly interpreted. The following **validation checklist** is applied to each recovered quotation:

#	Validation Check	Method	Result (hypothetical)
1	Image integrity	Re-verify image hash before VSC mounting	Match
2	Source traceability	Record VSC ID, path, MFT record no. of recovered file	VSC-3, D:\Tender\ LabPC2026\, MFT #48211
3	File signature	Header %PDF-1.7 and EOF marker present; opens without error	Verified
4	Content integrity	Hash of recovered file vs. hash of attachment in Vendor A's original email	SHA-256 identical
5	Metadata consistency	PDF creation date, author, $SI/$FN times consistent with timeline	Verified
6	Tool cross-validation	Recover again with a second tool (e.g., ShadowExplorer/libvshadow + Autopsy)	Same hash
7	Corroboration	Deletion confirmed by $UsnJrnl and M365 audit log	Verified
8	Peer review	Second examiner repeats key steps from notes	Verified
9	Documentation	Screenshots, tool versions, commands, hash values in appendix	Verified

Table 4.4: Validation checklist for recovered artefacts

Q30. Possible sources of error and alternative explanations

- **Timestamp interpretation:** NTFS "Accessed" times are often disabled or updated lazily; antivirus scans can change access times; copying changes Created time. These must not be read as user actions.
- **VSC timing:** a snapshot shows the state at creation time only; changes between snapshots are invisible, so the exact deletion time relies on $UsnJrnl, not on VSCs.
- **Automated deletion:** OneDrive sync conflicts or a cleanup policy can delete or move files without user action; the M365 audit log must show the "ClientIP" and user agent of the delete.
- **Shared credentials:** another employee could have used P's logged-on session (no screen lock). Check logon sessions, CCTV and attendance.
- **Legitimate reasons:** the clerk may claim the files were duplicates or withdrawn bids; check vendor communications.
- **Tool errors:** parsers may misinterpret corrupted records; cross-tool validation is necessary.
- **Clock and time-zone issues:** OneDrive logs are in UTC; workstation in IST; office server may have drift.
- **Partial recovery:** a file recovered from unallocated space (instead of a VSC) may be partially overwritten; only complete, hash-verified files are reported as findings.

Assignment 2(5): Multi-Source Investigation of a Suspected Cyber Incident

Case Scenario: "Ghost Login" (Case REC-CDF-2026-0729)

On **29 July 2026**, the college's online learning platform **RECLearn** (a cloud-hosted LMS) flagged that student account **2419XX045** (owned by "Student K") had logged in at **01:47 on 27 July** from an unusual IP, opened the faculty-only "Mid-Term Quiz Bank" through a misconfigured link, downloaded the answer key and changed quiz submissions of two other students. Student K denies involvement and says their password may have been reset without their knowledge.

With proper authorisation, investigators have access to Student K's **Windows 11 laptop**, **Chrome browser artefacts**, **Gmail** messages, **WhatsApp** records, an **Android phone (Samsung Galaxy M34)**, and **RECLearn/Google Workspace sign-in logs** and campus NAT logs.

Assumptions

- Student K consented to examination of the laptop and phone; cloud logs were obtained from the college IT administrator (the account provider).
- RECLearn logs record UTC timestamps; IST = UTC + 05:30.

Q31. Potential evidence from Windows and other operating systems

Source	Artefact	What it can reveal
Windows 11 laptop	Security.evtx (4624/4634/4800 lock/4801 unlock)	Whether the laptop was in use and unlocked at 01:47 on 27-07
	SRUM database (SRUDB.dat)	Per-application network usage per hour – was Chrome sending data at 01:00–02:00?
	Prefetch / Amcache / BAM	Execution of browsers, VPN clients (e.g., a free VPN), or remote-access tools (AnyDesk)
	Registry: WLAN profiles, NetworkList	Which Wi-Fi network (hostel, home, mobile hotspot) the laptop used
	LNK / Jump Lists / Downloads folder	Whether "MidTerm_AnswerKey.pdf" was downloaded or opened
	Windows Timeline / ActivitiesCache.db	Applications and documents in focus at the time
Android phone (Android 14)	SMS/OTP messages, notification log, app usage (UsageStats), Wi-Fi/location history	Whether the password-reset OTP reached the phone and whether the phone was in use
Other OS (if found)	Linux live-USB traces / macOS unified logs on another device	Use of another system to access the portal

Q32. What browser artefacts reveal and why careful interpretation is needed

Chrome artefacts are SQLite databases and files under %LOCALAPPDATA%\Google\Chrome\User Data\<Profile>\: **History** (urls, visits, downloads, keyword searches), **Cookies** (session cookies for reclearn.rec.edu – fictitious), **Login Data** (saved usernames, encrypted

passwords), **Web Data** (autofill), **Cache**, **Top Sites**, **Sessions/Tabs**, and **Preferences** (sync account, extensions). They may reveal visits to the quiz bank URL, a download of the answer key, the time of password-reset page visits and whether Chrome Sync was on.

Careful interpretation is necessary because:

- A URL entry may be created by a redirect, prefetch, or a link preview, not a deliberate visit; visit "transition type" must be checked.
- **Chrome Sync** can bring history from another device, a visit in the laptop's History may have occurred on a different device logged in with the same Google account.
- Timestamps are stored in WebKit format (microseconds since 1601-01-01 UTC) and must be converted correctly.
- Incognito browsing leaves almost no history; absence of records does not prove absence of activity.
- Browser extensions or malware can make requests on behalf of the user; a stolen session cookie lets an attacker act as the user from elsewhere.
- A shared browser profile does not identify which person was at the keyboard.

Q33. Evidence from email and messaging systems and correlation

System	Evidence types	Correlation possibilities
Email (Gmail – student account)	Full message headers (Received chain, Message-ID, originating IP, DKIM/SPF results), password-reset notifications from RECLearn, "new sign-in" security alerts, attachments, forwarding rules, filters that auto-delete alerts	Time of reset email ↔ browser visit to reset page ↔ RECLearn log; a hidden filter deleting alerts indicates account compromise by a third party
Messaging (WhatsApp on phone, WhatsApp Web on laptop)	msgstore.db (messages, timestamps, sender/receiver), media files (screenshot of answer key), group chats, linked-device list, call logs	Screenshot file hash ↔ downloaded PDF; WhatsApp Web session active on laptop at 01:50 ↔ Windows unlock event; who first shared the answer key
Telegram / Instagram DMs (if any)	Cloud-stored chats, channel posts of the answer key	Earliest distribution point; links to other students

Q34. Forensic considerations and challenges for the mobile phone

Considerations:

- **Isolation:** place the phone in airplane mode or a Faraday bag immediately to prevent remote wipe, new messages overwriting data, or cloud sync changes.
- **Keep powered and charged:** an unlocked phone in "After First Unlock" state gives far more access than a rebooted one; record battery level.
- **Document the state:** photograph screen, notifications, IMEI (*#06#), SIM ICCID, model, Android version.
- **Acquisition levels:** manual (screenshots), logical (ADB backup / vendor tools), file-system (full file system via tools like Cellebrite/Magnet AXIOM/Oxygen), physical (chip-off/JTAG) – choose the least intrusive level that answers the questions.
- **Consent / legal scope:** the phone contains extensive private data; examine only categories authorised (SMS, WhatsApp, browser, app usage for 25–28 July).

Challenges: full-disk/file-based encryption and screen locks; frequent OS and app updates break tool support; vendor diversity (Samsung, Xiaomi, etc.); WhatsApp end-to-end encryption and encrypted backups; data split between device, SIM and cloud; auto-deletion (disappearing messages); anti-forensic apps; very limited tools for physical acquisition of modern Android devices.

Q35. Internet and cloud-based evidence sources

Source	Holder	Relevant data
RECLearn LMS audit logs	College IT / LMS vendor	Login time, IP, user-agent, session ID, quiz-bank access, download events, submission edits
Google Workspace (rec.edu) sign-in & admin logs	College IT admin	Password reset events, recovery email/phone changes, 2-step verification status, login locations
Campus firewall / NAT logs	College Network team	Map internal private IP + port to public IP used at 01:47; link to hostel Wi-Fi MAC address
ISP subscriber records	Telecom/ISP (via police request under law)	Identify subscriber of an external IP (e.g., mobile data IP)
Google account data (Takeout/My Activity)	Google (with user consent/legal process)	Chrome Sync history across devices, location history, device list
WhatsApp / Meta records	Meta (legal process only)	Subscriber info, linked-device information (no message content)
Code/file sharing sites (Drive, Telegram)	Service providers	Upload time and uploader of the answer key

Q36. Cross-source correlation table

Figure 5.1 shows how evidence sources connect to the incident; Table 5.3 correlates specific hypothetical artefacts.

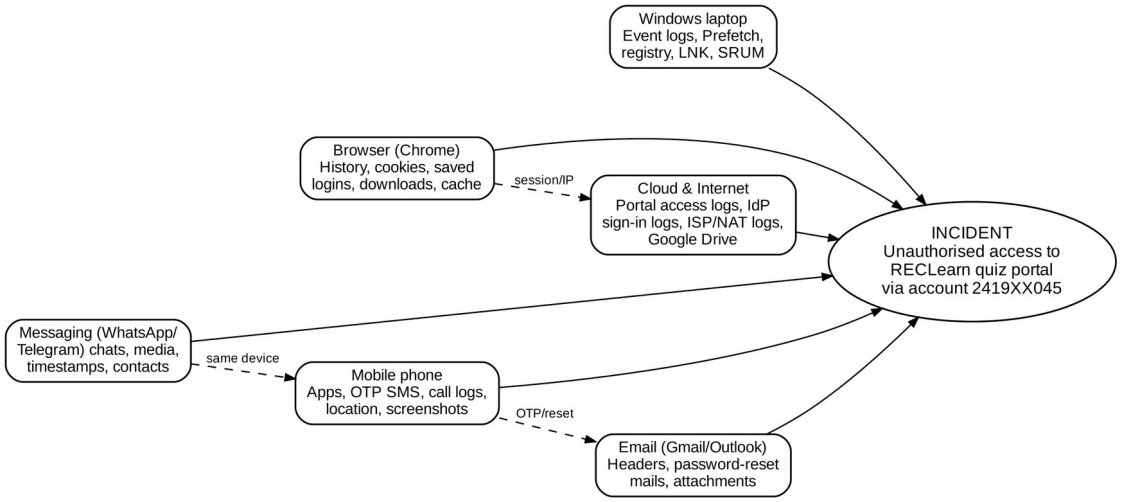


Figure 5.1: Multi-source evidence map for Case REC-CDF-2026-0729

#	Time (IST) 27-07-2026	Computer (Windows)	Browser	Email / Messaging	Mobile	Cloud / Internet	Inference
1	01:31	Laptop locked since 23:58 (4800), no 4801 until 07:12	–	Gmail: “Password reset requested” to K’s student mail 01:31	No OTP SMS received; phone screen off (UsageStats)	Google Admin log: reset via recovery email changed on 20-07	Reset not performed from K’s laptop/phone
2	01:33	–	–	Gmail filter created 20-07 that auto-archives “security alert” mails	–	Admin log: filter created from IP 49.xx.xx.12 (mobile ISP)	Account likely already compromised on 20-07
3	01:47	No network activity in SRUM for Chrome 01:00–02:00	No RECLearn visits in local History	–	Phone on hostel Wi-Fi idle	RECLearn login 2419XX045 from 49.xx.xx.12, UA “Chrome/126 Linux”	Login from a different device and network
4	01:52	–	Chrome Sync shows visit to quiz-bank URL from device “Linux-PC”	–	–	LMS: download of MidTerm_AnswerKey.pdf; submissions of 2 students edited	Attacker device synced to K’s Google account
5	08:15	Laptop: K opens WhatsApp Web	–	WhatsApp group “CSE-B Squad”: answer-key screenshot posted by Student R at 02:10	K’s phone receives the screenshot 08:15 (first seen)	Telegram channel post of same file 02:25	K received the file after distribution; R is a lead
6	20-07, 16:05	No saved RECLearn password in laptop “Login Data”	History: K logged in to RECLearn on a shared library PC (Sync device “LIB-PC-08”)	–	–	Library PC session not logged out	Possible credential theft from shared library PC

Table 5.3: Cross-source correlation matrix

Q37. Major challenges when evidence is distributed across devices and cloud

#	Challenge	Explanation and impact in this case
1	Jurisdiction & legal access	Google, Meta and the LMS vendor may store data outside India; obtaining records requires legal process (MLAT/letters rogatory) or provider cooperation and takes time.
2	Volume & variety of data	Laptop, phone and cloud produce hundreds of GB in different formats; correlating them requires specialised tools and time.
3	Encryption	Android FBE, WhatsApp end-to-end encryption, BitLocker and HTTPS hide content; lawful key/password access is often necessary.
4	Time synchronisation	UTC in cloud logs, IST on devices, WebKit/Unix epoch formats, device clock drift – errors of minutes can break correlation.
5	Volatility & retention	Cloud logs (e.g., 30-day retention), NAT logs and disappearing messages may expire before preservation requests are made.
6	Attribution	IP addresses behind CGNAT/mobile networks are shared by many users; Sync and shared accounts break the link between account and person.
7	Anti-forensics	Email filters hiding alerts, incognito mode, VPNs and wiping apps reduce evidence.
8	Tool limitations & validation	Rapid app updates change database schemas; different tools may parse differently, cross-validation needed.
9	Privacy & proportionality	Phones hold intimate personal data of the student and others; examinations must be narrowly scoped under DPDP Act principles.
10	Chain of custody for cloud data	Cloud exports are not “seized” physically; integrity must be shown by hashing exports, documenting the export method and provider certificates.

Q38. Final investigation conclusion

Category	Findings
Direct evidence	• RECLearn logs show account 2419XX045 logged in at 01:47 IST on 27-07-2026 from public IP 49.xx.xx.12 using a Linux Chrome user-agent, accessed the quiz bank, downloaded the answer key and edited two submissions. • Google Workspace admin logs show the password was reset at 01:31 via a recovery email address that had been changed on 20-07-2026 from the same IP 49.xx.xx.12.
Corroborating evidence	• Student K’s laptop was locked from 23:58 to 07:12 with no Chrome network activity (Security log, SRUM). • K’s phone received no OTP and was idle on hostel Wi-Fi. • Chrome Sync records show a foreign device “Linux-PC” and a library PC “LIB-PC-08” linked to K’s Google account. • The answer-key screenshot was first posted to WhatsApp by Student R at 02:10 and reached K only at 08:15.
Unresolved questions	• Who is the subscriber of IP 49.xx.xx.12 at 01:31–01:52? (requires ISP request through police) • Who owns the device “Linux-PC”? Is it linked to Student R? • How was K’s password first obtained, the un-logged-out library PC session on 20-07 or phishing? • Was the misconfigured faculty link shared intentionally by anyone?

Category	Findings
Overall conclusion	The available evidence indicates that the account 2419XX045 was used in the incident but does not support the conclusion that Student K personally carried it out; it is more consistent with account takeover after credential exposure on 20-07-2026. Student R and the IP 49.xx.xx.12 are priority leads. Recommendations: enforce MFA on RECLearn, auto-logout on library PCs, fix the faculty link permissions, and preserve all logs immediately.

Conclusion

The five case studies in this mini project applied the CR23532 syllabus to realistic, campus-based scenarios. Across all of them, the same forensic principles recur: act only with authorisation; preserve the original and work on hash-verified copies; maintain an unbroken chain of custody; normalise and document time; correlate independent artefacts before drawing conclusions; separate the actions of an **account** from the actions of a **person**; and report findings, limitations and alternative explanations honestly, in language the deciding authority can understand.

The assignments also showed how specialised tools fit into the workflow, IXImager and the ASB container for acquisition and preservation, ILookIX for analysis, and Volume Shadow Copy analysis for recovering earlier states of evidence, and how modern investigations increasingly depend on cloud logs and mobile devices, bringing challenges of jurisdiction, encryption, volatility and privacy.

References

1. CR23532 – Cyber and Digital Forensics, Course Syllabus and Unit Notes, Rajalakshmi Engineering College.
2. B. Nelson, A. Phillips and C. Steuart, *Guide to Computer Forensics and Investigations*, 6th ed., Cengage Learning, 2019.
3. B. Carrier, *File System Forensic Analysis*, Addison-Wesley, 2005.
4. H. Carvey, *Windows Forensic Analysis Toolkit*, 4th ed., Syngress, 2014.
5. NIST SP 800-86, *Guide to Integrating Forensic Techniques into Incident Response*, 2006.
6. ISO/IEC 27037:2012, *Guidelines for identification, collection, acquisition and preservation of digital evidence*.
7. Perlustro, ILookIX and IXImager product documentation.
8. The Information Technology Act, 2000 (as amended); Bharatiya Sakshya Adhiniyam, 2023; Digital Personal Data Protection Act, 2023 – Government of India.

Note: all scenarios, names, identifiers and timestamps in this report are hypothetical and created by the author for academic purposes.